

FILTER DESIGN TEST CASE

Duplexer Module - n78 (3.3-3.8 GHz 5G NR) for GPS/GNSS Anti-Jamming Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Duplexer Module filter targeting n78 (3.3-3.8 GHz 5G NR) for GPS/GNSS Anti-Jamming Filter. The filter is designed on PZT on Si substrate with a target center frequency of 2211.4 MHz and bandwidth of 356.7 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the GPS/GNSS Anti-Jamming Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Duplexer Module
- Frequency Band: n78 (3.3-3.8 GHz 5G NR)
- Application: GPS/GNSS Anti-Jamming Filter
- Substrate: PZT on Si
- Package: Fan-Out WLP
- Design Tool: Cadence AWR Microwave Office
- Design Center: Singapore RF Lab

This specification applies to all variants of the Duplexer Module design targeting n78 (3.3-3.8 GHz 5G NR). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	2211.4 MHz
Bandwidth (3 dB):	356.7 MHz
Insertion Loss Target:	≤ 1.9 dB
Return Loss Target:	≥ 8.0 dB
Out-of-Band Rejection:	≥ 29.0 dB
Isolation (Duplexer):	≥ 22.0 dB
Q Factor:	7788
Temperature Coefficient:	-15.9 ppm/°C
Die Size:	0.8 x 2.0 mm
Wafer Size:	6-inch
Number of Resonators:	4
Electrode Material:	Al

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open Cadence AWR Microwave Office and load the Duplexer Module template project.
2. Define the substrate stack: PZT on Si with specified layer thicknesses.
3. Set design targets: center freq 2211.4 MHz, BW 356.7 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.9 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Fan-Out WLP).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, PZT on Si).
10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 6634 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.9 dB
- Return loss in passband shall be >= 8.0 dB
- Out-of-band rejection at +/- 535 MHz offset >= 29.0 dB
- Group delay variation across passband shall be <= 3 ns
- Temperature drift over -40°C to +85°C shall be <= 4.4 MHz
- Power handling shall be >= +29 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 94%

6. TEST RESULTS

Measurement Date: 2024-04-02
Devices Tested: 73
Insertion Loss (meas): 1.55 dB
Return Loss (meas): 10.5 dB
Rejection (meas): 31.6 dB
Isolation (meas): 26.8 dB
Q Factor (meas): 7788
Group Delay Variation: 3.3 ns
Power Handling: +27 dBm
Die Yield: 97.3%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

- a) The Duplexer Module design demonstrated below-target performance for n78 (3.3-3.8 GHz 5G NR).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Package parasitic effects were consistent with 3D EM model predictions.
- e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

- 1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

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